

Introduction to probability

Probability spaces · Live writing edition

Soma S Dhavala · IIT Jammu

2026-09-27

Errors in a test sample

Suppose a fixed classifier makes 8 errors on 100 tested items. What has been measured?

A large grid of dots for taking notes.

A probability question about the same sample

A test sample has 100 items, 8 in error. If one item is selected uniformly, what is the probability it is an error?

A statistical question about future items

Do the 8 errors in the specified sample of 100 establish a future-item error probability of 0.08? State the inference assumption needed.

Random selection of a chord

A chord is selected in a circle. What must be specified before its length can be assigned a probability?

An inspection experiment

Inspect two components in order. Record P for pass and F for fail. List the possible records.

An event as a set of outcomes

For the ordered inspection outcomes PP, PF, FP, FF, list the event A: at least one component fails.

Exactly one failure

For outcomes PP, PF, FP, FF, let A be at least one failure. Find the event B of exactly one failure and compare it with A.

Intersection: both conditions

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cap D$.

Union: at least one condition

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cup D$.

Complement: the remaining outcomes

For $\Omega=\{PP,PF,FP,FF\}$, let $A=\{PF,FP,FF\}$ be at least one failure. Find A^c .

Disjoint events

For the ordered inspection outcomes, let $B=\{PF,FP\}$ be exactly one failure and $E=\{FF\}$ be two failures. Determine whether they are disjoint.

A large grid of dots for writing the answer.

Practice: two alarm indicators

Record 1 when an alarm is active, 0 otherwise, and use order for two indicators. List the outcomes in “at least one active,” “exactly one active,” and “neither.”

Assigning probabilities to outcomes

Suppose the ordered inspection outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Check that these are valid outcome probabilities.

Adding the probabilities in an event

The ordered outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Calculate the probability of exactly one failure.

Using the complement

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Calculate the probability of at least one failure directly and by the complement.

A large grid of dots for writing the solution, consisting of 20 rows and 50 columns of small, light gray dots.

Deriving the complement rule

Explain why $P(A^c) = 1 - P(A)$.

A large grid of dots for writing the explanation. The grid consists of 20 rows and 100 columns of small, light blue dots, providing a structured space for the user to write their answer.

Overlapping events in the inspection model

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Let C be first-component failure and D second-component failure. Find $P(C)$, $P(D)$ and $P(C \cap D)$.

A large grid of dots for working space.

Deriving the addition rule

In the component-inspection model, C is first-component failure and D second-component failure. Use $P(C)=0.20$, $P(D)=0.10$ and $P(C \cap D)=0.02$ to find $P(C \cup D)$.

Practice: overlapping system alerts

Assume $P(A) = 0.20$, $P(B) = 0.15$ and $P(A \cap B) = 0.08$ for two system alerts. Find the probability of any alert.

Practice: exactly one system alert

Assume $P(A)=0.20$, $P(B)=0.15$ and $P(A \cap B)=0.08$ for two system alerts. Calculate the probability of exactly one alert.

Equally likely outcomes

A batch contains 20 items, of which 3 are defective. Select one item uniformly. Find the defect probability.

A die with unequal probabilities

Suppose a die has $P(\{k\}) = k/21$ for $k = 1, \dots, 6$. Find the probability of an even face.

Grid area for writing the solution.

Practice: checking a proposed model

Three mutually exclusive outcomes exhaust an experiment. Proposed probabilities are 0.2, 0.3 and 0.6. Is the model valid?

The sample space in a complete model

For one inspection, record 0 for pass and 1 for failure. Specify the possible outcomes.

The collection of events

For the inspection recorded as 0 = pass and 1 = failure, list every subset of $\Omega = \{0, 1\}$.

Open the interactive experiment

The probability assignment

Suppose a pass/fail inspection has failure probability 0.10. Using 0 = pass and 1 = failure, assign a probability to every event.

The probability triplet

The example has outcomes $0 = \text{pass}$ and $1 = \text{failure}$, all four events, and failure probability 0.10 . Write the probability-space triplet.

A large grid of dots for writing the probability-space triplet. The grid consists of 20 rows and 100 columns of small, light gray dots, providing a space for the user to write their answer.

Why the event collection must be closed

If we can ask whether A occurs, what related questions should the model also describe?

A large rectangular area filled with a grid of small, light blue dots, intended for taking notes or writing.

The probability rules

Complements let us ask what cannot occur; countable unions let us ask whether at least one of a list of events occurs. Once these questions are events, what rules should their probabilities obey?

A large rectangular area filled with a grid of small, light gray dots, intended for taking notes or writing.

Running frequency in a model

In the independent fair-coin model, the lab plots the running fraction of heads beside the model probability p . What can a finite run illustrate? What does it establish about a limit?

[Open the interactive experiment](#)



Practice: a batch inspection triplet

Ten labelled items contain two defective items, numbered 3 and 8. Select one item uniformly. Specify the triplet and the defect event.

Recording a numerical result

For the inspection model, the probabilities of PP, PF, FP and FF are 0.72, 0.08, 0.18 and 0.02. Let X count failed components. Assign X to each outcome.

The event $X = 1$

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find $P(X=1)$.

A large grid of dots for writing the solution, consisting of 20 rows and 50 columns of small, light blue dots.

The distribution of the failure count

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find its probability distribution.

A large grid of dots for writing the solution.

A numerical set and its inverse image

For PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. For $B=\{1,2\}$, find $X^{-1}(B)$ and $P(X \in B)$.

An inverse image need not be an inverse function

On $\Omega=\{PP,PF,FP,FF\}$, X counts failures. Find $X^{-1}(\{1\})$ and explain whether an outcome can be recovered uniquely from $X=1$.

A random variable and a probability measure

For outcomes PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. Find $X(\text{PF})$ and $P(\{\text{PF}, \text{FP}\})$. What does each map take as input?

Practice: a repair-cost random variable

For the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$, let X count failures and $Y=500X$ rupees. Find $P(Y>500)$.

A uniform measurement model

Let E be a uniform measurement error on $[-2, 2]$ mm. State how to calculate the probability of an interval $[a, b]$ contained in this range.

Meeting a tolerance

Assume measurement error E is uniform on $[-2, 2]$ mm. Find the probability that the absolute error is at most 0.5 mm.

A large grid of dots for working space, consisting of 20 rows and 50 columns of small, light gray dots.

A uniform source variable

Let U be uniform on $(0,1)$. Find $P(U < 0.25)$ and $P(U > 2/3)$.

Transforming a uniform variable

Let U be uniform on $(0,1)$ and $D=10U$ cm. Find $P(D < 3 \text{ cm})$.

A large grid of dots for writing the solution, consisting of 20 rows and 50 columns of small, light gray dots.

A nonlinear transformation

Let U be uniform on $(0,1)$ and $D=10\sqrt{U}$ cm. Find $P(D<3 \text{ cm})$.

The chord-length question

For a circle of radius R , find the probability that a sampled chord has length $L > \sqrt{3}R$. What sampling rule is missing from the question?

[Open the interactive experiment](#)



Chord length from the midpoint distance

Let D be the distance from the centre to the chord. Express its length L in terms of D .

A large grid of dots for writing the answer.

Converting the long-chord condition

Using $L=2\sqrt{(R^2-D^2)}$, express the strict condition $L>\sqrt{3}R$ as an inequality for D .

A numerical geometry check

For a circle of radius $R=10$ cm and $L=2\sqrt{(R^2-D^2)}$, calculate L when $D=3$ cm and when $D=6$ cm. Compare both lengths with $\sqrt{3}R$.

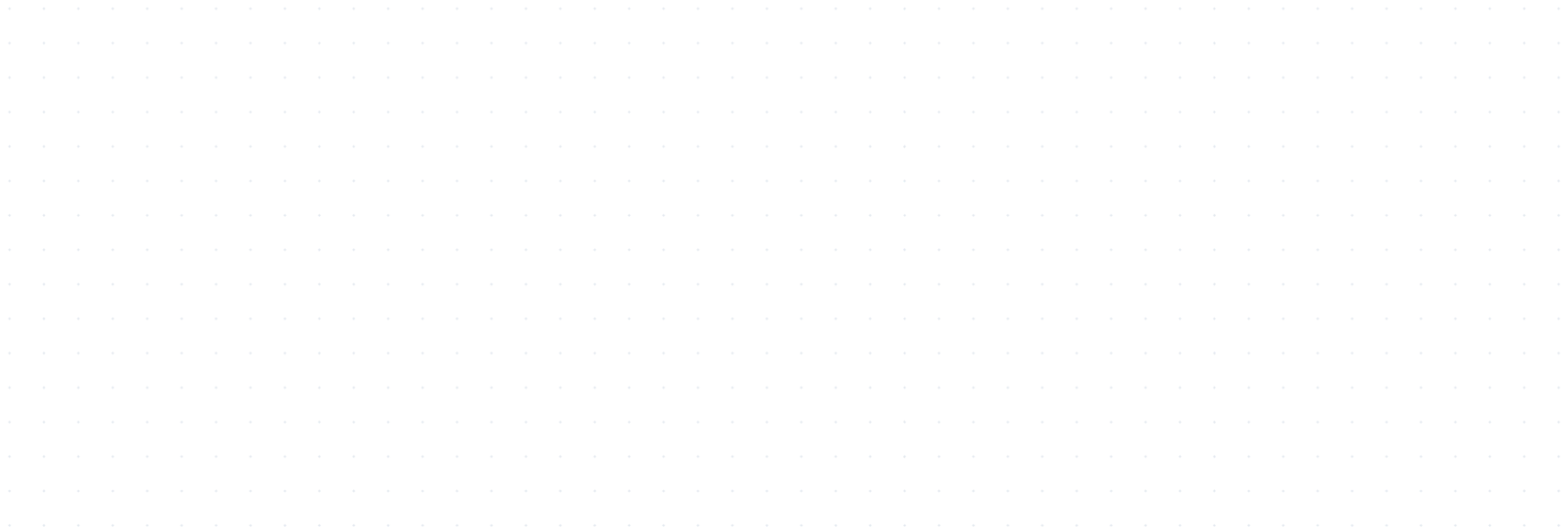
Coordinates that specify a chord

For a circle with radius R , let $t=D/R\in(0,1)$ and $\phi\in[0,2\pi)$ be the direction of the chord midpoint. Let $C(\phi,t)$ denote the chord through the point $R t(\cos\phi,\sin\phi)$, perpendicular to that radius.



Recovering the chord coordinates

For a chord with midpoint m in a circle of radius R , define $t=||m||/R$ and $\phi=\text{Arg}(m)$ in $[0,2\pi)$. Recover (ϕ,t) from the chord.



Model B: uniform radial distance

Sample D uniformly on $(0,R)$ and an independent uniform direction ϕ . For $L=2\sqrt{(R^2-D^2)}$, find the probability of $L>\sqrt{3}R$.



Model B: sampling map and inverse

Let U, V be independent uniform variables on $(0,1)$ and $[0,1)$, and set $t=U$, $\phi=2\pi V$. Using $C(\phi, t)$, write the chord, recover (U, V) , and express its length.

A large grid of dots for writing the answer, consisting of 20 rows and 100 columns of small, light gray dots.

Model C: a midpoint uniform in disk area

Choose the chord midpoint uniformly by area in a disk of radius R . The long-chord event is $D < R/2$. Find the probability.



Why uniform area uses a square root

For a midpoint sampled uniformly by area in a radius- R disk and U uniform on $(0,1)$, derive $P(D < r)$ and the corresponding transformation of U .

A large grid of dots for taking notes, consisting of 20 rows and 100 columns of small, light gray dots.

Model C: sampling map and inverse

For uniform disk area, let $t=\sqrt{U}$ and $\phi=2\pi V$, where U,V are independent uniform variables on $(0,1)$ and $[0,1)$. Using $C(\phi,t)$, write the chord, its inverse and its length.

A large grid of dots for writing the answer, consisting of 20 rows and 100 columns of small, light gray dots.

Practice: a smaller midpoint region

For the same disk of radius R , find $P(D < R/3)$ under (i) a uniform radial distance and (ii) a midpoint chosen uniformly by disk area.

Model A: two uniform endpoints

Choose independent uniform endpoint angles A_1, A_2 on $[0, 2\pi)$, and let $\Theta = (A_2 - A_1) \bmod 2\pi$. Express the chord length using Θ .

A large grid of dots for writing the answer, consisting of 20 rows and 100 columns of small, light gray dots.

Model A: the favourable angular interval

Let $\Theta = (A_2 - A_1) \bmod 2\pi$ be uniform on $[0, 2\pi)$ and $L = 2R \sin(\Theta/2)$. Solve the strict inequality $L > \sqrt{3}R$ for Θ .

A large grid of dots for solving the inequality. The grid consists of 20 rows and 100 columns of small, light blue dots, providing a workspace for the student to work out the solution.

Model A in midpoint coordinates

For a uniform relative angle Θ on $[0, 2\pi)$, define $\alpha = \min(\Theta, 2\pi - \Theta)$. Find the distribution of α and the chord midpoint distance $t = D/R$.



Model A: sampling map and inverse

For independent uniform endpoint angles, put $U=\alpha/\pi$, where $\alpha=\min(\Theta, 2\pi-\Theta)$, and let $V=\phi/(2\pi)$ be the normalized midpoint direction. With $t=\cos(\alpha/2)$, find the chord map and its inverse.



Model A: the length random variable

Under independent uniform endpoint angles, Θ is uniform on $[0, 2\pi)$, $\alpha = \min(\Theta, 2\pi - \Theta)$, $U = \alpha/\pi$ and $t = \cos(\alpha/2)$. Express $L = 2R\sqrt{1-t^2}$ in terms of U ; find the condition for $L > \sqrt{3}R$.

A large grid of dots for working space, consisting of 20 rows and 100 columns of small, light gray dots.

Practice: recovering the source coordinates

For $t=1/2$ and midpoint angle $\phi=\pi$, recover $v=\phi/(2\pi)$ and u under Model A, $t=U$; Model B, $t=U$; and Model C, $t=\sqrt{U}$. Is the chord longer than $\sqrt{3}R$?

Comparing the three sampling distributions

Summarize the event calculations using the same uniform source U .

Resolution of Bertrand's paradox

Why are the three probabilities compatible?

A large grid of dots for taking notes.

Changing coordinates in one fixed model

Keep Model B, where $T = D/R$ is uniform on $(0, 1)$. Write $W = T^2$. What is $P(W < 1/4)$?

A large grid of dots for writing the answer.

Sampling records from two sites

A hypothetical dataset has 80 records from site A and 20 from site B. Select one of the 100 records uniformly. Find the probability that it came from each site.

Selecting a site before selecting a record

A hypothetical dataset has 80 records from site A and 20 from site B. Select A or B with probability $1/2$ each, then uniformly within the selected site. Find the selection probability of one specified record from each site.

Accuracy and class proportions

A hypothetical test set contains 9,900 nondefective and 100 defective items. A system always predicts “nondefective.” Calculate its accuracy and defect detection fraction.

Practice: two fair dice

Assume two independent fair dice. List the 36 equally likely ordered outcomes and find the probability that the sum is 7.

Practice: at least one six

Assume two independent fair dice. Using the complement, find the probability of at least one six.

Review: describing a probability calculation

A hypothetical log has 50 requests, 7 above 200 ms. Select one request uniformly. Define the outcome, the latency random variable, and the event of latency above 200 ms; calculate its probability.

Review: assumptions behind the answer

For a probability calculation based on data or a transformed random variable, what must be specified about the experiment, event and distribution?

Further reading

Use the main notes for additional worked problems and the supplement for the formal development.